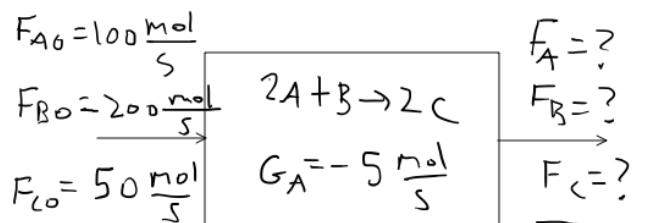


01 Deriving Mole Balances for Common Reactors

Let's start with a "warm up" example from Material and Energy Balances:



The reaction $2A + B \rightarrow 2C$ is being carried out in a reactor. A, B, and C are fed at rates of 100 mol/s, 200 mol/s, and 50 mol/s, respectively. $G_A = -5$ mol A/s. Determine F_C .

Accumulation = In - Out + net generation

assume steady state

write the balance on species C

$$0 = F_{C0} - F_C + G_C$$

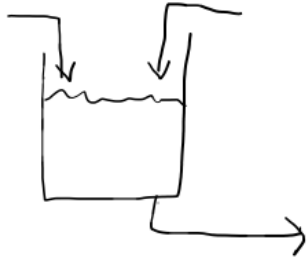
$$F_C = F_{C0} + G_C$$

$$F_C = 50 \frac{\text{mol}}{\text{s}} + (G_A) \left(\frac{\nu_C}{\nu_A} \right) = 50 \frac{\text{mol}}{\text{s}} + \left(-5 \frac{\text{mol A}}{\text{s}} \right) \left(\frac{2 \text{ mol C}}{-2 \text{ mol A}} \right)$$

$F_C = 55 \frac{\text{mol}}{\text{s}}$

Let's derive the mole balance for the CSTR.

- What is a CSTR and how does a CSTR work? [University of Michigan Equipment Encyclopedia: CSTR](#)



- CSTR Key Assumptions:

- One: Well mixed
- Two: Steady state
- Three: CSTR operates at the exit conditions

- Taking these into account, the balance becomes:

$$0 = F_{A0} - F_A + G_A \rightarrow \text{since the CSTR is well m}$$
$$G_A = r_A V$$

$$0 = F_{A0} - F_A + r_A V$$

- What variable do we solve for?

$$V = \frac{F_{A0} - F_A}{-r_A}$$

Example: The liquid phase reaction $2A \rightarrow B$ is being carried out isothermally in a CSTR. The rate law is kC_A^2 and the rate constant equals $0.03 \text{ L/mol}\cdot\text{s}$. Species A is fed to the reactor at a concentration of 2 M and a flowrate of 3 L/s . Species A exits the reactor at a concentration of 0.1 M . Determine:

- The rate of reaction in the CSTR:

$$r_A = -kC_A^2$$

$$r_A = -\left(0.03 \frac{\text{L}}{\text{mol}\cdot\text{s}}\right)\left(0.1 \frac{\text{mol}}{\text{L}}\right)^2$$

$$r_A = -0.0003 \frac{\text{mol}}{\text{L}\cdot\text{s}}$$

- The volume of the CSTR:

$$V = \frac{F_{A0} - F_A}{-kC_A^2} \quad C = \frac{F}{v} \leftarrow \text{script } v = \text{volumetric flowrate}$$

$$V = \frac{v(C_{A0} - C_A)}{-kC_A^2} = \frac{\left(3 \frac{\text{L}}{\text{s}}\right)\left(2 \frac{\text{mol}}{\text{L}} - 0.1 \frac{\text{mol}}{\text{L}}\right)}{0.0003 \frac{\text{mol}}{\text{L}\cdot\text{s}}}$$

$V = 19000 \text{ L}$

Now let's derive the balance for the batch reactor:

- What is a batch reactor and how does a batch reactor work? [University of Michigan Equipment Encyclopedia: Batch reactor](#)

- 1) fill the batch reactor
- 2) mix it up so that it's well mixed
- 3) carry out the reaction
- 4) drain
- 5) clean

- Batch reactor key assumptions:

- One: well mixed

- Two: no in flows or out flows

- Taking these into account, the balance becomes:

$$\frac{dN_A}{dt} = r_A V$$

- Special case: Constant volume batch reactor:

$$\frac{dC_A}{dt} = r_A$$

Let's do the same example as above, but this time for a batch reactor.

- Reaction: $2A \rightarrow B$
- Reactor: Isothermal, constant volume batch reactor
- Rate law: kC_A^2 , $k = 0.03 \text{ L/mol}\cdot\text{s}$.
- Initial concentration of A = 2 M.

- What variable do we solve for in the batch reactor?

time or reaction time

- Determine: The batch time needed to achieve $C_A = 0.1 \text{ M}$.
- The rate of reaction in the batch reactor is a function of time

- Applying and solving the batch reactor mole balance:

$$\frac{dC_A}{dt} = r_A$$

$$r_A = -k C_A^2$$

$$\frac{dC_A}{dt} = -k C_A^2$$

$$\int_{C_{A0}=2\text{M}}^{C_A=0.1\text{M}} \frac{dC_A}{C_A^2} = -k \int_0^t dt$$

$$-\frac{1}{0.1} + \frac{1}{2} = -kt$$

$$\boxed{t = 317 \text{ s}}$$

Let's derive the mole balance for the PFR.

- What is a PFR and how does a PFR work? [University of Michigan Equipment Encyclopedia: PFR](#)

tubular reactor, turbulent flow, also called "plug flow" as the fluid flows down the length of the reactor reactants are converted to products.

- PFR Key Assumptions:

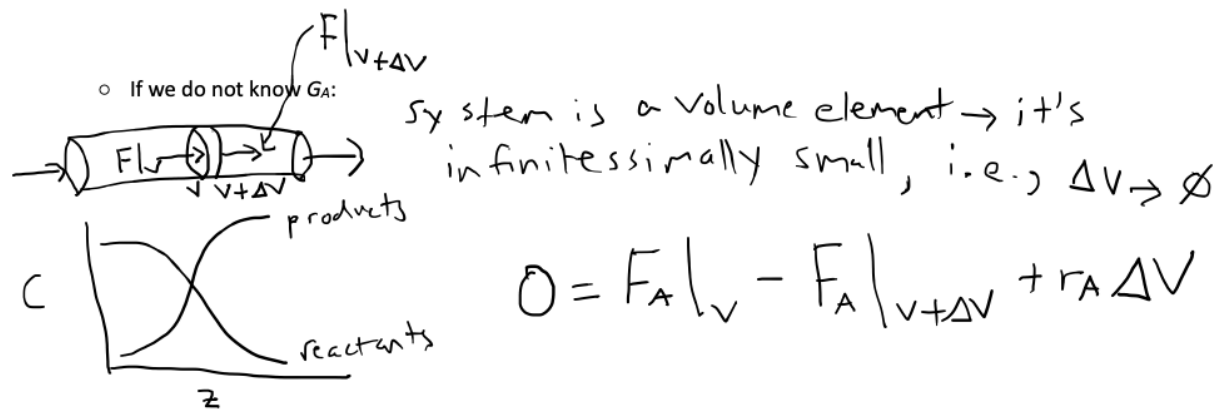
- One: steady state
- Two: perfect mixing in the radial dimension
- Three: continuous variation of the concentration (and temperature) down the length of the PFR

- Like for the batch reactor, the concentration (and temperature) is not constant in the PFR

- How do we define the system?  $G_A = \text{some value}$

- If we know G_A :

$$0 = \dot{Q}_{A, in} - \dot{F}_A + G_A$$



Apply the definition of the derivative. Divide by ΔV and take the limit as $\Delta V \rightarrow 0$.
(This works perfectly with our assumption that ΔV is infinitesimally small.) Then:

$$\lim_{\Delta V \rightarrow 0} \frac{F_A|_V - F_A|_{V+\Delta V}}{\Delta V} = -r_A$$

$$= -\frac{dF_A}{dV}$$

$$\frac{dF_A}{dV} = r_A$$

- If we do not know G_A :

Apply the definition of the derivative. Divide by ΔV and take the limit as $\Delta V \rightarrow 0$.
(This works perfectly with our assumption that ΔV is infinitesimally small.) Then:

